Endong Dai

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Profile

Aspiring hardware engineer specializing in **RTL design, FPGA systems, and digital IC development**, with experience in **Verilog, C/C++, PLECS, and MATLAB simulation**. Strong foundation in **power electronics and embedded system**, backed by research in **AI chip design** and **floating-point ALU implementation**. Passionate about creating efficient, high-performance hardware architectures bridging digital logic and system integration.

EDUCATION

Duke University, Durham, NC — M.S. in ECE (Hardware Track) — Aug 2025 – Present
University of Nottingham (China + UK 2+2) — B.Eng. in EEE, First Class (GPA 3.8/4.0) — Sep 2021 – Jul 2025
Awards: Dean's Scholarship (Year 1); Third Prize – Lego EV3 Robot Programming Competition

Skills

- **Hardware Description & Design:** Verilog (Quartus, Vivado), Cadence (Virtuoso)
- **Circuit & Simulation Tools:** PLECS, LTspice, MATLAB/Simulink, KiCad, STM32CubeIDE
- **Programming & Embedded Systems:** C/C++, Assembly (PIC16, MIPS), Arduino, Raspberry Pi
- **Software & Platforms:** Linux(Ubuntu), ROS, Git

RESEARCH & PROJECT EXPERIENCE

Diffusion Chip Design (Gather-Scatter Unit)– Duke University *Sep 2025 – Jan 2026*

- Implementing a Gather/Scatter Unit (SGU) for **irregular data** of a Diffusion AI chip, designing a 4-stage **pipelined** FSM (Index fetch, PISO, SRC read, DST write) to stream H×W×C INT8 **feature maps** over AXI while following **ready/valid handshake protocol**.
- Design logic that supports **gather** (index-driven random SRC reads, contiguous DST writes) and **scatter** (contiguous SRC reads, index-driven random DST writes), with base addresses configured via AXI registers.
- Writing **Verilog RTL** of SGU module and maintaining the project on **GitHub** with **VS code**.

16-bit Floating-Point ALU Design (Add, Subtract, Multiply, Divide) - Duke University *Sep 2025 – Dec 2025*

- Designed an **IEEE 754 half-precision ALU (binary16)** supporting four arithmetic operations.
- Implemented **12×12-bit array multiplier** and **Restoring divider (accurate to 5 decimal places)** for high-accuracy floating-point computation.
- Developed **exception detection, normalization, rounding**, and **top-level integration** modules in Verilog.
- Estimated post-layout metrics (TSMC 65 nm): **16 MHz max, < 18 mW, 0.008 mm² area**.

Omni-Directional Robot Collision Awareness - University of Nottingham Ningbo China *Jun 2024 – Aug 2024*

- Configured **ROS Noetic** on Ubuntu 20.04 with **Gazebo** simulation and **RViz** visualization for robot navigation.
- Deployed **Cartographer SLAM** algorithm using LiDAR to map multi-room environments; integrated **RoboRTS-Base** navigation stack for 2D goal tracking.

Design and Development of a Two-Switch Forward DC/DC Converter - University of Nottingham *Feb 2024 – Apr 2024*

- Designed and simulated a **Two-Switch Forward Converter** using **PLECS** to achieve a regulated 10 V DC output.
- Created a **two-layer PCB** in **KiCad**, integrating MOSFETs, transformer, and filtering network.
- Conducted **open-loop and closed-loop simulations** and verified control performance with **MATLAB**.
- Assembled the converter and validated stable operation under varying loads during final demonstration.

Development of a Heart Rate and Pulse Oximeter Device - University of Nottingham . *Oct 2023 – Dec 2023*

- Implemented **SpO_2 and heart rate detection** using STM32 microcontroller with dual LED driver and photodiode sensing.
- Designed **analog front-end circuits** (TIA, low/high-pass filters, dual-stage amplifier) and implemented FFT-based heart rate extraction.

- Developed **ADC firmware** for signal digitization and applied biomedical formulas for SpO_2 calculation.